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BASIC BLOCK

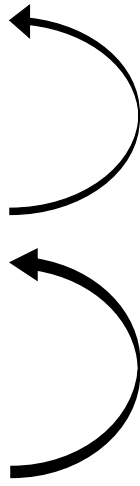
How to select LEADER

1. The first three address instruction in the intermediate code is a leader.
2. Any instruction that is the target of a conditional or unconditional jump is a leader.
3. Any instruction that immediately follows of a conditional or unconditional jump is a leader.

Consider this example to find leader &
divide into blocks:

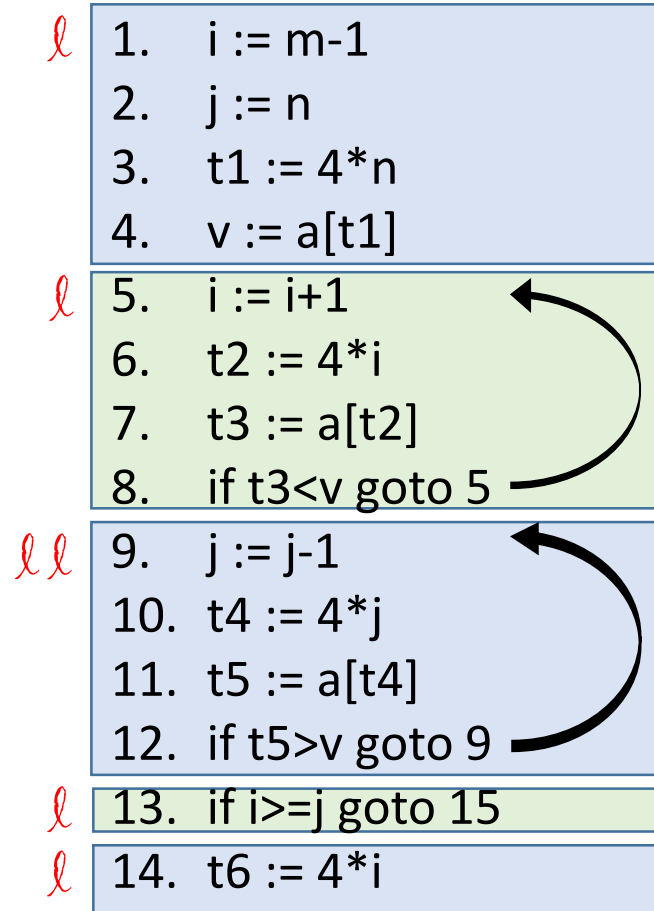
1. $i := m-1$
2. $j := n$
3. $t1 := 4*n$
4. $v := a[t1]$
5. $i := i+1$
6. $t2 := 4*i$
7. $t3 := a[t2]$
8. if $t3 < v$ goto 5
9. $j := j-1$
10. $t4 := 4*j$
11. $t5 := a[t4]$
12. if $t5 > v$ goto 9
13. if $i \geq j$ goto 15
14. $t6 := 4*i$

ℓ 1. $i := m-1$
2. $j := n$
3. $t1 := 4*n$
4. $v := a[t1]$
ℓ 5. $i := i+1$
6. $t2 := 4*i$
7. $t3 := a[t2]$
8. if $t3 < v$ goto 5
ℓℓ 9. $j := j-1$
10. $t4 := 4*j$
11. $t5 := a[t4]$
12. if $t5 > v$ goto 9
ℓ 13. if $i \geq j$ goto 15
ℓ 14. $t6 := 4*i$



First, we find leader (*ℓ*)

- The first three address instruction in the intermediate code is a leader
- Any instruction that is the target of a conditional or unconditional jump is a leader
- Any instruction that immediately follows of a conditional or unconditional jump is a leader.



First, we find leader (ℓ)

- The first three address instruction in the intermediate code is a leader
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Thank You